

CLAUDE BUILDER CLUB @ VANDERBILT

Claude AI Hackathon

Participant Guide

Welcome, Vandy Builders!

Let's reimagine what is possible through transformative innovations and big ideas. This hackathon is your chance to use Claude AI to tackle some of humanity's biggest challenges and to show what is possible when talented people build with purpose.

The Hackathon Theme: Social Impact

Anthropic CEO Dario Amodei wrote an essay titled "Machines of Loving Grace" envisioning a future where AI helps eliminate disease, expand opportunity, strengthen democracy, and help people find meaning. As Claude Builder Club members, you have the opportunity to play a role in making this future a reality.

This hackathon's theme is social impact. You will choose from four tracks, each focused on a different area where AI can make a meaningful difference. Your projects should demonstrate how AI could genuinely contribute to human flourishing, while wrestling with the hard ethical questions along the way.

We want to see your thinking as much as your code.

What We're Looking For

Across all tracks, judges will be looking for projects that embody these principles:

- **Solutions to real problems for specific people** - not hypothetical problems for "everyone"
- **AI that empowers people** rather than replacing or diminishing them
- **Deep thought about ethics** - who benefits, who might be harmed, and what safeguards are needed
- **Working prototypes** that demonstrate core functionality
- **Clear understanding** of both the problem you're solving and your approach

Three Questions Every Team Must Answer

1. **Who are you building this for and why do they need it?**
2. **What could go wrong and what would you do about it?**
3. **How does this help people rather than make decisions for them?**

Getting Started

Here's a suggested approach to kick off your project:

4. **Pick your track** based on your expertise and what you're passionate about
5. **Define the specific problem** - who faces it and why it matters
6. **Build a working prototype** that demonstrates your core idea
7. **Test your assumptions** - talk to potential users if possible
8. **Think through risks** and build in safeguards
9. **Prepare to explain your thinking** as much as your code

The Four Tracks

Choose the track that aligns with your skills and passions. Each track focuses on a different area of social impact.

1. Physical & Mental Health

Building tools and systems that improve access to care, support wellbeing, and bridge gaps in physical and mental health services for individuals and communities.

2. Finance & Economic Systems

Building tools that improve financial access, model economic realities, and create more equitable systems - from personal finance to macroeconomic policy.

3. Educational Empowerment

Creating solutions that democratize learning, close equity gaps, and give every person the tools and knowledge to grow and thrive.

4. Creative Flourishing

Exploring how technology and human creativity intersect to unlock new forms of expression and heightened creative freedom.

Track 1: Physical & Mental Health

How can you improve access to care, help people understand and manage their health, or reduce the barriers that keep people from getting support either physically or mentally?

The Problem: Healthcare is expensive, inaccessible, and confusing. Medical knowledge that could help people is locked behind jargon and paywalls. Billions lack access to quality care, and even those with access struggle to navigate complex systems. Mental health is even further behind because it is stigmatised, underfunded, and deeply inaccessible for most of the world.

Example Project Ideas

- Diagnostic aids for underserved clinics
- Patient education platforms that translate medical jargon into plain language
- Medication adherence and reminder tools
- Symptom assessment and triage helpers
- Mental health check-in and early intervention tools
- Peer support platforms with safety guardrails
- Stress, burnout, and resilience tools for high-risk communities
- Accessibility tools for people with disabilities navigating healthcare

Ethical Considerations to Address

- Medical accuracy - when should your tool hand off to a real professional?
- The particular sensitivity of mental health data and the risk of harm from poor responses
- Privacy of deeply sensitive health data
- Bias in medical AI across different populations and health presentations
- Transparency about limitations and when professional care is needed

Track 2: Finance & Economic Systems

How can you help people understand money, build economic resilience, and navigate financial systems that were often not designed with them in mind?

The Problem: Financial systems are complex, opaque, and frequently exclusionary. Predatory products target those with least, while wealth-building knowledge flows to those who already have it. Billions of people make high-stakes financial decisions about debt, savings, housing, and business, without access to trustworthy, personalised guidance.

Example Project Ideas

- Financial literacy platforms for underserved or unbanked communities
- Tools that demystify debt, interest, and credit for first-time borrowers
- Budget and cash flow tools adapted for irregular or informal incomes
- Small business financial planning aids for entrepreneurs without accountants
- Explainers for tax systems, benefits, and entitlements people are missing out on
- Tools that model the long-term impact of everyday financial decisions
- Fraud and predatory lending detection and awareness tools

Ethical Considerations to Address

- The high stakes of financial advice - bad guidance can cause lasting harm
- Who gets excluded by your design (language, literacy, internet access)?
- Empowering informed decisions vs. nudging people toward particular choices
- Avoiding the reinforcement of existing inequalities through biased models
- Transparency about the limits of automated financial guidance vs. regulated advice

Track 3: Educational Empowerment

How can you make quality learning more accessible, more personalised, and more relevant to people's real lives and goals?

The Problem: Where you're born still largely determines what you get to learn and become. Quality education is gatekept by geography, income, language, and time. Millions of motivated people, such as students, career changers, and self-taught learners, are trying to grow without the resources, feedback, or guidance that others take for granted.

Example Project Ideas

- Adaptive tutoring systems that meet learners where they are
- Career transition tools that map skills to new opportunities
- Job interview preparation and confidence-building aids
- Tools that support learners with different abilities or learning styles
- Practical language learning grounded in real-world contexts
- Scholarship and opportunity matching platforms
- Mentorship-gap fillers for first-generation students

Ethical Considerations to Address

- Who gets excluded by your design - tech requirements, literacy levels, language?
- The risk of automating bad advice when guiding real career or education decisions
- Empowering learners to think and grow vs. creating dependency on AI shortcuts
- Cultural assumptions baked into definitions of "success" or "good learning"
- Data privacy for children and vulnerable learners

Track 4: Creative Flourishing

How can you amplify human creativity, help people find meaning, or support cultural expression and preservation?

The Problem: As AI handles more routine work, questions about human purpose become urgent. Access to creative development and cultural preservation isn't equal. People who could find meaning through creative expression often don't get the chance.

Example Project Ideas

- Community storytelling platforms
- Music or art education with real-time feedback
- Writing development tools
- Language preservation projects
- Skill development for crafts or instruments
- Intergenerational knowledge exchange
- Purpose exploration guides

Ethical Considerations to Address

- Keeping humans as creators using tools, not AI as creator with human prompter
- Enhancing vs. short-cutting the creative process in ways that diminish it
- Cultural appropriation vs. respectful celebration
- Homogenisation of creative voices
- Creating vs. destroying value for creative workers

How You'll Be Judged

Your project will be evaluated across four categories, totalling 100 points. Understanding these criteria will help you focus your efforts and deliver the strongest possible presentation.

Impact Potential (25 points)

Does your project solve a real problem? Who benefits and how? Could it scale?

Score	What It Looks Like
21–25	Addresses a specific, meaningful problem with a clear population in mind. It's immediately obvious who benefits and why this matters. The solution directly addresses the problem in a way that makes sense.
13–20	Addresses a real problem but the population or benefit is somewhat vague. The connection between solution and problem is logical but could be stronger.
6–12	The problem is loosely defined or the solution doesn't clearly address it. Hard to tell who this is for or why they'd use it.
0–5	The problem is unclear or the solution has little connection to it.

Technical Execution (30 points)

Does the core functionality work? Does it demonstrate the idea effectively? Is AI used purposefully?

Score	What It Looks Like
25–30	Core functionality works and effectively demonstrates the idea. AI is used in a way that feels purposeful and well-considered. You can see how it would work for a real user.
16–24	Core functionality mostly works. AI is used adequately. Some gaps in the demo but the concept comes through clearly.
6–15	Significant gaps in functionality or the demo doesn't clearly show how it works. AI use feels superficial or tacked on.
0–5	Doesn't work or is a mockup only. Little evidence of meaningful technical implementation.

Ethical Alignment (25 points)

Did your team think seriously about potential harms? Does your project empower people rather than replace them?

Score	What It Looks Like
21–25	The team clearly wrestled with ethical implications. Can speak to who might be harmed, what could go wrong, and what they've done about it. The solution puts people in control. Ethical reasoning feels genuine, not performative.
13–20	The team considered some ethical implications and had reasonable answers. The solution is broadly empowering. Some gaps in ethical reasoning but shows genuine awareness.
6–12	Ethical considerations feel like an afterthought. The team struggles to articulate potential harms or safeguards.
0–5	No meaningful ethical consideration. Solution risks disempowering or harming users with no acknowledgment.

Presentation (20 points)

Can your team clearly explain what you built, why you built it, and what you'd do next?

Score	What It Looks Like
17–20	Clear, confident explanation of the problem, solution, and impact. Demo shows the product working. The team can articulate next steps and handle questions well.
11–16	Covers the key points. The demo is functional. The team communicates the core idea clearly even if the presentation is rough around the edges.
5–10	Hard to follow or key pieces are missing. The demo is unclear or incomplete. Struggles to answer questions about the project.
0–4	Incoherent or incomplete. Can't demonstrate functionality or explain what was built.

Hackathon Schedule

Time	Event
10:00 – 10:30 AM	Check-in
10:30 – 11:00 AM	Introductions, Schedule Overview, Judging Criteria & Project Tracks
11:00 AM – 12:00 PM	Workshop: <i>"My Secret Superpower? Skills. How they work, why they work, and how to build best of class."</i> Speaker: Dr. Jesse Spencer-Smith, Chief Data Scientist & Interim Director of the College of Connected Computing
12:00 – 1:00 PM	Hacking begins
1:00 – 2:00 PM	Lunch
2:00 PM – 4:00 PM	Hacking continues
4:00 PM – 5:30 PM	Presentations & judging
5:30 PM – 5:45 PM	Judges convene
5:45 PM – 6:00 PM	Results

Helpful Resources

Make the most of these Anthropic resources to get building quickly:

- **Developer Account & API Keys** - set up [here](#)
- **Quickstart Guide** - make your first [API call](#)
- **Claude Code** - get started with [agentic coding](#) from your terminal
- **Anthropic Cookbook** - code snippets and guides for [common workflows](#)
- **API Documentation** - [full reference](#) for all features and products

Important: Set Up Your API Credits Before the Event

Make sure you've attended a Claude Builder Club event and completed the Builder's Club check-in form with the correct ID to receive your \$25 in API credits. Do this well before hackathon day so you're ready to build from the start. API credits are processed on Friday afternoons.

Good luck, and happy building!

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